



ALLIANCE
UNIVERSITY

NAAC
GRADE A+
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ASSIGNMENT- 02

Bachelor of Computer Applications (Hons. Artificial Intelligence & Machine Learning)
Semester- III

Subject: Database Management System

Topic: ER Diagram Design and Relational Schema

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ER Diagram and Relational Schema

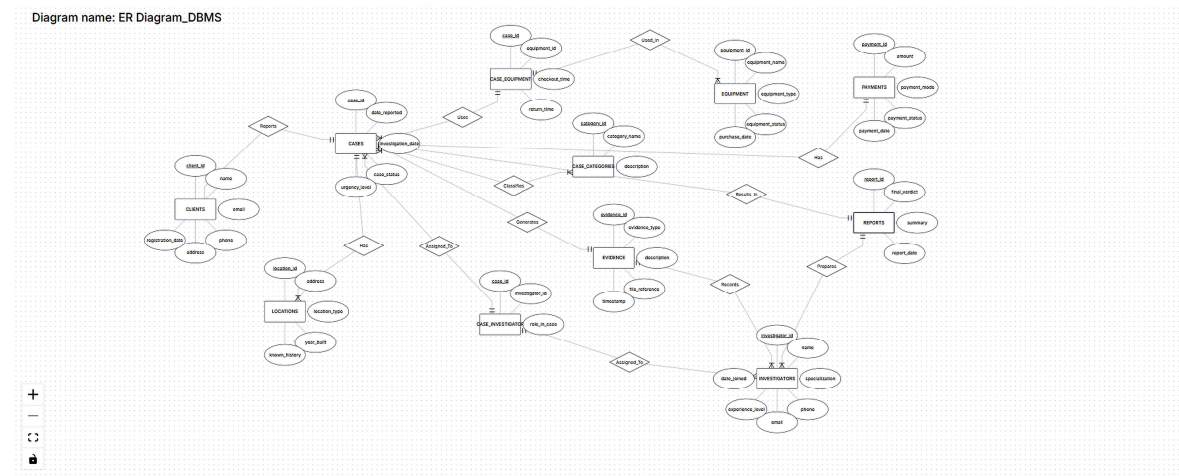
Application: Paranormal Investigation Agency Management System

1. List of Entities and Attributes

Primary keys are shown in bold. Foreign keys are marked (FK).

No.	Entity	Attributes
1	CLIENTS	client_id (PK), name, email, phone, address, registration_date
2	CASES	case_id (PK), client_id (FK), location_id (FK), category_id (FK), date_reported, investigation_date, case_status, urgency_level
3	LOCATIONS	location_id (PK), address, location_type, year_built, known_history
4	CASE_CATEGORIES	category_id (PK), category_name, description
5	INVESTIGATORS	investigator_id (PK), name, specialization, phone, email, experience_level, date_joined
6	CASE_INVESTIGATOR	case_id (PK, FK), investigator_id (PK, FK), role_in_case
7	EQUIPMENT	equipment_id (PK), equipment_name, equipment_type, condition_status, purchase_date
8	CASE_EQUIPMENT	case_id (PK, FK), equipment_id (PK, FK), checkout_time, return_time
9	EVIDENCE	evidence_id (PK), case_id (FK), recorded_by (FK), evidence_type, description, file_reference, timestamp
10	REPORTS	report_id (PK), case_id (FK), prepared_by (FK), final_verdict, summary, report_date
11	PAYMENTS	payment_id (PK), case_id (FK), amount, payment_mode, payment_status, payment_date

2. ER Diagram



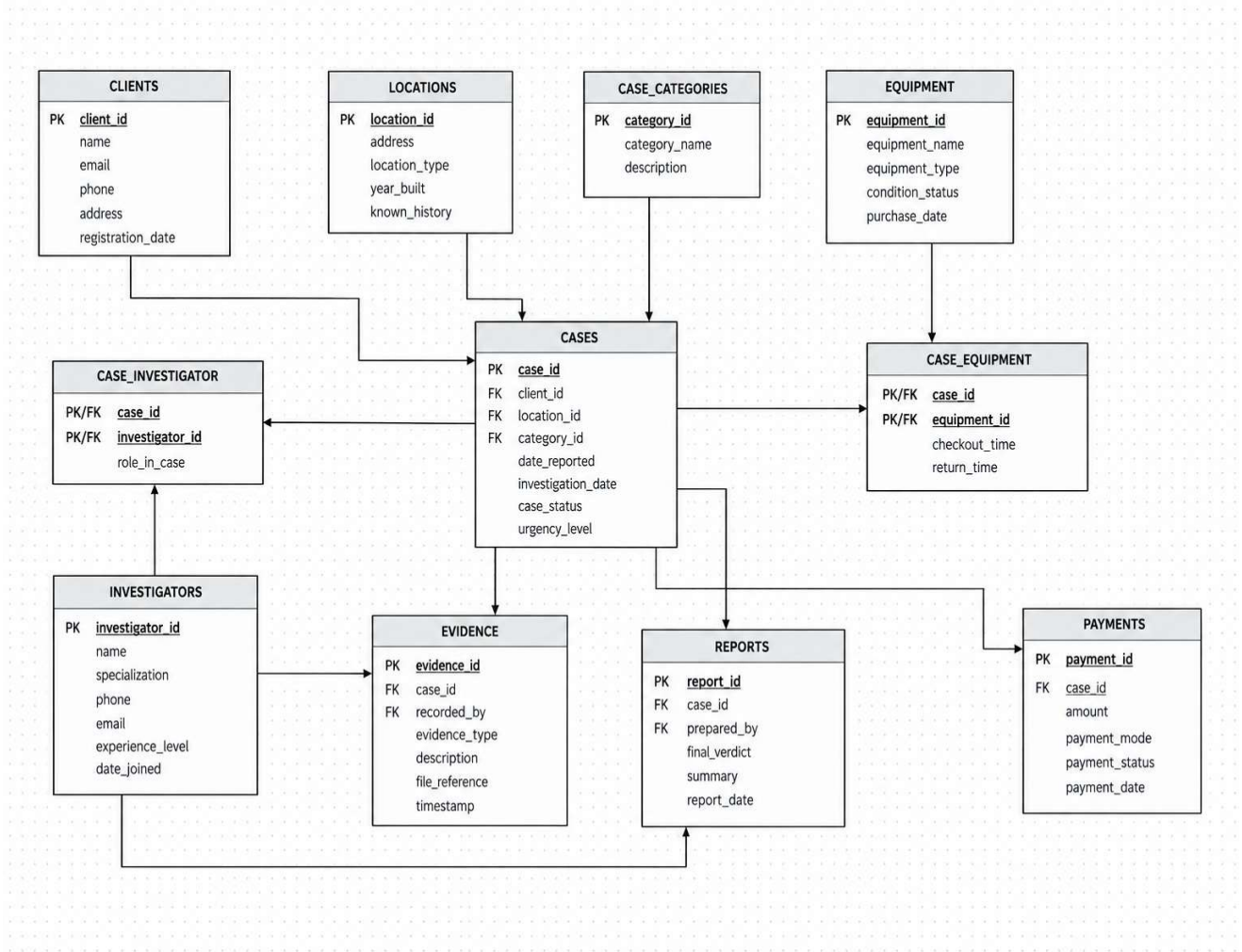
Notation:

- **rectangles** are entities,
- **ellipses** are attributes (underlined = primary key),
- **diamonds** are relationships,
- **crow's-foot** symbols show cardinality.

3. Relationships and Cardinalities

Relationship	Entity 1	Entity 2	Cardinality
REPORTS	CLIENTS	CASES	One-to-Many (1:M)
HAS	LOCATIONS	CASES	One-to-Many (1:M)
CLASSIFIES	CASE_CATEGORIES	CASES	One-to-Many (1:M)
ASSIGNED_TO	CASES	CASE_INVESTIGATOR	One-to-Many (1:M)
ASSIGNED_TO	INVESTIGATORS	CASE_INVESTIGATOR	One-to-Many (1:M)
USES	CASES	CASE_EQUIPMENT	One-to-Many (1:M)
USED_IN	EQUIPMENT	CASE_EQUIPMENT	One-to-Many (1:M)
GENERATES	CASES	EVIDENCE	One-to-Many (1:M)
RECORDS	INVESTIGATORS	EVIDENCE	One-to-Many (1:M)
RESULTS_IN	CASES	REPORTS	One-to-One (1:1)
PREPARES	INVESTIGATORS	REPORTS	One-to-Many (1:M)
HAS	CASES	PAYMENTS	One-to-One (1:1)

4. Relational Schema



5. Relations (Table)

Table	Attributes
CLIENTS	client_id , name, email, phone, address, registration_date
LOCATIONS	location_id , address, location_type, year_built, known_history
CASE_CATEGORIES	category_id , category_name, description
EQUIPMENT	equipment_id , equipment_name, equipment_type, condition_status, purchase_date
INVESTIGATORS	investigator_id , name, specialization, phone, email, experience_level, date_joined
CASES	case_id , client_id, location_id, category_id, date_reported, investigation_date, case_status, urgency_level
CASE_INVESTIGATOR	case_id , investigator_id , role_in_case
CASE_EQUIPMENT	case_id , equipment_id , checkout_time, return_time
EVIDENCE	evidence_id , case_id, recorded_by, evidence_type, description, file_reference, timestamp
REPORTS	report_id , case_id, prepared_by, final_verdict, summary, report_date
PAYMENTS	payment_id , case_id, amount, payment_mode, payment_status, payment_date

Bold attributes form the primary key of each table.

Conversion rules used

- Each entity became a table, and its key attribute became the primary key.
- For every 1:M relationship, the primary key of the “one” side was added as a foreign key in the “many” side table (for example, client_id, location_id and category_id in CASES).
- The 1:1 relationship between CASES and REPORTS, and between CASES and PAYMENTS, places case_id in REPORTS and in PAYMENTS as a foreign key.
- The M:N relationships between CASES and INVESTIGATORS, and between CASES and EQUIPMENT, became the associative tables CASE_INVESTIGATOR and CASE_EQUIPMENT, whose composite primary keys are made of the two respective foreign keys.

6. Primary and Foreign Key Identification

Table	Primary Key	Foreign Key	References
CLIENTS	client_id	–	–
LOCATIONS	location_id	–	–
CASE_CATEGORIES	category_id	–	–
EQUIPMENT	equipment_id	–	–
INVESTIGATORS	investigator_id	–	–
CASES	case_id	client_id location_id category_id	CLIENTS(client_id) LOCATIONS(location_id) CASE_CATEGORIES(category_id)
CASE_INVESTIGATOR	case_id, investigator_id (composite)	case_id investigator_id	CASES(case_id) INVESTIGATORS(investigator_id)
CASE_EQUIPMENT	case_id, equipment_id (composite)	case_id equipment_id	CASES(case_id) EQUIPMENT(equipment_id)
EVIDENCE	evidence_id	case_id recorded_by	CASES(case_id) INVESTIGATORS(investigator_id)
REPORTS	report_id	case_id prepared_by	CASES(case_id) INVESTIGATORS(investigator_id)
PAYMENTS	payment_id	case_id (unique)	CASES(case_id)

7. Brief Relationship Explanation

- **CLIENTS – CASES (Reports, 1:M):**

One client can report many cases over time, but each case is reported by exactly one client.

- **LOCATIONS – CASES (Has, 1:M):**

One location can be the subject of many cases, but each case takes place at exactly one location.

- **CASE_CATEGORIES – CASES (Classifies, 1:M):**

One category can apply to many cases, but each case belongs to exactly one category.

- **CASES – CASE_INVESTIGATOR (Assigned_To, 1:M):**

One case can have many entries in the junction table, one for each investigator assigned to it.

- **INVESTIGATORS – CASE_INVESTIGATOR (Assigned_To, 1:M):**

One investigator can appear in many junction records, since they may work on several cases.

- **CASES – CASE_EQUIPMENT (Uses, 1:M):**

One case can use several pieces of equipment, each recorded as a separate entry.

- **EQUIPMENT – CASE_EQUIPMENT (Used_In, 1:M):**

One piece of equipment can be checked out for many cases over time.

- **CASES – EVIDENCE (Generates, 1:M):**

One case can generate many pieces of evidence, but each evidence record belongs to exactly one case.

- **INVESTIGATORS – EVIDENCE (Records, 1:M):**

One investigator can record many pieces of evidence across different cases.

- **CASES – REPORTS (Results_In, 1:1):**

Each case results in exactly one final report, and each report belongs to exactly one case.

- **INVESTIGATORS – REPORTS (Prepares, 1:M):**

One investigator can prepare reports for many cases, but each report is prepared by one investigator.

- **CASES – PAYMENTS (Has, 1:1):**

Each case is billed through exactly one payment record.